

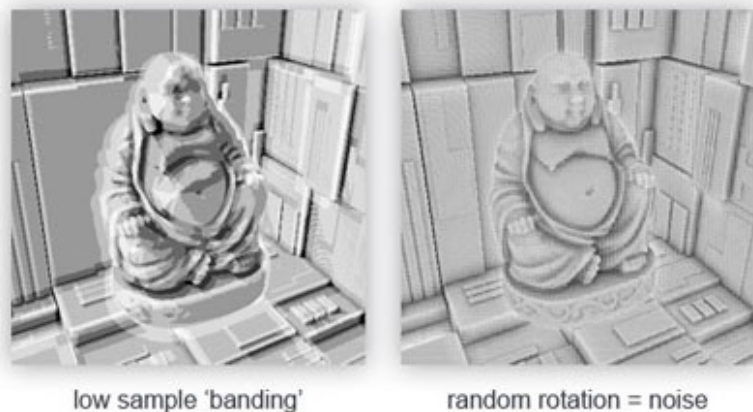
SSAO Assignment Guideline

3rd session: Improving SSAO

After finishing the previous sessions, you will be able to render something that is similar to ambient occlusion but with several artifacts. In this guideline, I will mention the improvements most used in the literature. However, you are required to include randomness and blurring in your work and you are free to implement those other extra that you prefer.

Adding randomness

One of the main problems of simple kernels is the banding effects artifact. It is caused because, as we are performing only a few samples, if we constantly sample in the same groups of close pixels won't sample the same surrounding parts.



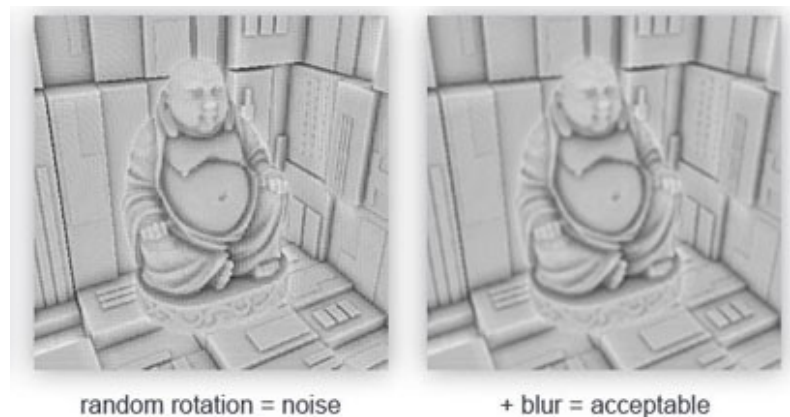
source: learnopengl.com

To avoid that, we need to force the samples to be different for every pixel. As computing random numbers are quite expensive, one solution is to generate a random texture. The idea is to use a texture where we store random numbers in each channel that we use to modify the sampling directions. There are several ways to do this: adding a rotation angle, adding a jittering in the sampling direction, rotating directly the sampling direction using a TBN matrix, ...

Removing the randomization noise

After including randomization in our sampling strategy, we are able to remove the banding artifacts, but at the price of introducing noise in our images. This noise is caused by the lack of samples that we need to use to be able to achieve real-time frame rates.

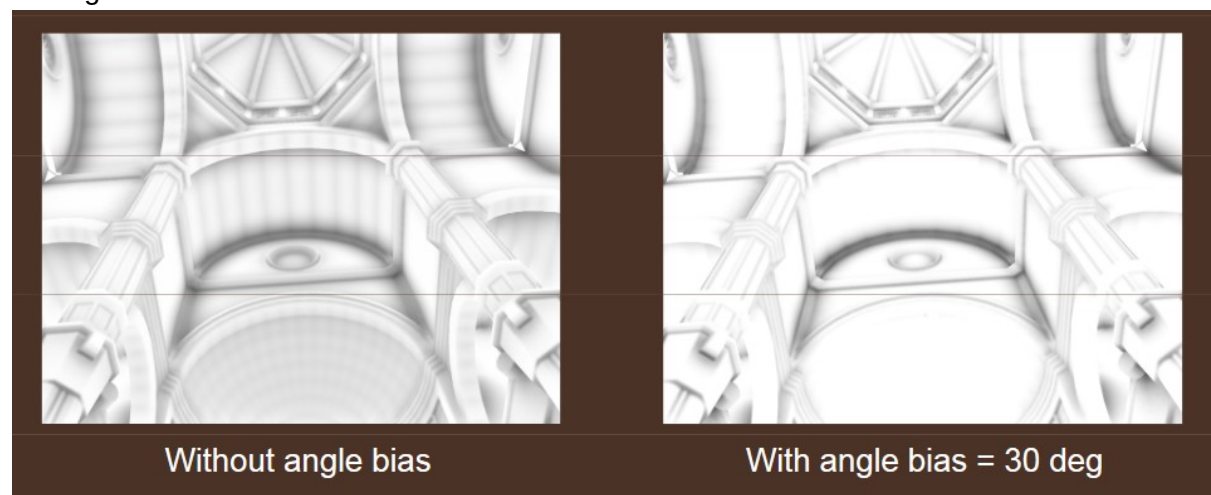
The most used solution to this is adding a third rendering step where we apply a blurring process to the AO layer. Several kernel options can be used: Gaussian filters, bilateral filters, etc. You can implement some of these and discuss the pros and cons of them.



source: learnopengl.com

Removing occlusion of nearly tangent surfaces

Nearly tangent surfaces can influence the result of the ambient occlusion technique by adding more occlusion than is desired:



source: slides of HBAO in SIGGRAPH 2008

One way to avoid this overestimation of the AO term is to add a bias angle that will define the angle from which we consider a sample is not tangent to the surface.

Range Check

One simple improvement is to avoid that distant surfaces contribute to the ambient occlusion term. This can be done by adding a radius that we can consider as a distance threshold to decide when a sample is considered or not. Related to that, other techniques use attenuation factors to handle the discontinuities between the sampled surfaces.



without range check



with range check

source: learnopengl.com

Compute AO in Half-Resolution

Since ambient occlusion is mostly low frequency one possible acceleration can be computing the ambient occlusion in half resolution. In this case, the blurring step still have to be done in full resolution in order to avoid bleeding across edges.

Final render

After you have computed the final AO image, the final rendering will only need to combine it with the color image of your scene. This combination can be done directly by multiplying the final color/shaded color of your fragment with its AO term.